Sai Abhishek Avusula

Senior Azure Data Engineer

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PROFESSIONAL SUMMARY:

- **11+** years of IT experience in various tools & technologies, excelling in requirement gathering, Data Modeling, Analysis, ETL Development, Validation, Deployment, Monitoring, and Visualization reports.
- **5+** years of experience in implementation of **Azure Cloud Components- Azure Data Factory, Azure Data Lake (ADLS Gen 1 & Gen 2), Azure Blob storage, Azure Databricks, Synapse Analytics, Stream Analytics, HDinsight, Azure SQL DB/DW, Azure Cosmos DB, Azure SQL Server Alteryx & Power BI.**
- Specialized in architecting and deploying **Azure Data Factory** (V2) pipelines, scheduling automated **triggers**, mapping intricate **data flows**, and securely handling credentials via **Azure Key Vault**. sOptimized query performance and data modeling using **Dimensional Modeling** techniques, ensuring structured and efficient reporting.
- Developed and optimized **Apache Spark Databricks notebooks**, employing **Python** (**PySpark**) and **SparkSQL** to execute complex data transformation pipelines, which facilitated the structured transition of datasets within **Azure Data Lake Storage Gen2 (ADLS Gen 2)** from raw ingestion to strategically curated zones, ensuring data readiness for downstream analytics and machine learning applications.
- Implemented CI/CD pipelines for Azure-based data solutions using **Azure DevOps**, enhancing automation and improving deployment cycles for data platforms. Automated messaging workflows leveraging **Azure Service Bus** to ensure reliable enterprise data integration.
- Experience in migration of on-premises databases to **Microsoft Azure environment (Blobs, Azure Data Warehouse, Azure SQL Server, PowerShell Azure components, SSIS Azure components).**
- Proficient in Apache Spark operations, with a solid understanding of resilient distributed datasets (**RDD**s), **DataFrames**, and **Datasets**, encompassing transformations, actions, and data persistence techniques for **advanced analytics**.
- Enforced robust security measures in **Snowflake**, including role-based access control, data encryption (both at rest and in transit), and network policies to safeguard sensitive information. Managed authentication and access control with **Azure Active Directory (AAD)** for enhanced identity management.
- Designed and implemented scalable data warehousing solutions using **Snowflake**, leveraging **SnowSQL**, **Zero-Copy Cloning**, and **Time Travel** to optimize performance, reduce costs, and ensure data versioning and rollback capabilities.
- Proficient in master data management using **Azure Purview**, with expertise in **ETL processes**, **data security**, **data governance**, **data profiling**, ensuring **data quality**, and **Matillion** for data extraction, transformation, and loading, executing sophisticated data modeling strategies using **Star Schema** and **Snowflake Schema** methodologies.
- Skilled in orchestrating multi-container applications using **Docker Compose**, ensuring seamless service integration and network configuration.
- Strong knowledge of **Dimensional Modeling techniques**, including **Star Schema, Snowflake Schema**, **Kimball**, and **Inmon** methodologies for data warehousing
- Developed robust real-time data streaming pipelines using **Azure Notification Hubs**, allowing for event-driven push notifications to be delivered efficiently across distributed systems.
- Experienced in deploying and managing applications using **Docker**, optimizing Dockerfile configurations for efficiency and security.
- Integrated **Kubernetes** with CI/CD pipelines for automated deployment processes, leveraging tools like **Jenkins** and **GitLab** CI.
- Extensive experience working with large-scale financial data, ensuring compliance, data security, and governance in industries such as Banking, Investment, and Insurance.
- Demonstrated expertise in deploying and scaling applications with **Kubernetes**, managing workloads with Deployments, Services, and Ingress controllers.
- Automated **container** deployment and management tasks using **Bash** and **Python** scripts, improving efficiency and reducing manual errors.

- Utilized Snowflake's advanced features, such as Materialized Views, Time Travel, and Automatic Clustering, to optimize data storage and access, reducing costs and improving **data analysis** readiness.
- Integrated **Azure Data Catalog** to streamline data discovery and metadata management, ensuring effective governance and searchability across enterprise data assets.
- Developed and optimized **HQL (Hive Query Language)** queries within the Hadoop ecosystem, improving batch processing efficiency and enabling seamless data transformations for analytical workloads.
- Highly proficient in orchestrating complex workflows using tools like **Oozie**, **Airflow**, **Terraform**, and **Azure Data Factory** to streamline data pipeline processes.
- Demonstrated expertise in the **Hadoop** ecosystem, proficient in managing **HDFS** and **Yarn**, orchestrating data processing with **MapReduce**, and developing complex analytics with **Apache Spark**.
- Possess deep expertise in **PySpark**, with a proven track record of engineering robust **Spark applications** in **Python** for scalable data processing and analysis.
- Proficient in data ingestion using **Sqoop** and **Flume**; cluster coordination with **Zookeeper**; data management and querying with **Hive**, **Pig**, and **HBase**; event streaming with **Kafka**; and efficient workflow scheduling with **Oozie**.
- Proficient with big data transformation tools such as **Apache NiFi** and **Databricks Delta** for advanced data wrangling, ensuring efficient ETL operations within big data ecosystems.
- Good hands-on experience in **Spark Core**, **SparkSQL**, **Scala**, and **Spark Streaming**, and Implemented **Snowflake architecture** which stored and analyzed all data records in one place.
- Have extensive experience in connecting AWS resources like S3 bucket, RDS, and Redshift and creating pipelines to move data from AWS to Azure.
- Proficient in Data Analytics by using various tools, **Tableau, Alteryx, Power BI, Azure Synapse Analytics, SSRS, Alteryx,** and **Microsoft Excel.** Created reports, dashboards, data models, dataset analysis, and administration of **Power BI** tools and QLIKVIEW.
- Experience in developing **Dashboard and Parameterized Reports using SSRS, Tableau,** and **Power BI**.
- Experience in **RDBMS** concepts such as Tables, User Defined Data Types, Indexes, Indexed Views, Functions, Table Variables and Stored Procedures.
- Transformed raw data of all forms – structured, semi-structured, and unstructured – using my expertise in **ETL**, data modeling, and analytics to craft actionable insights.
- Experience in working with **ETL/ELT/EDL tools** for developing workflows to extract, transform, and load data into different database systems.
- Hands-on experience with **DBT Labs** for **data transformation and modeling**, enabling efficient ELT processes, **source-to-target mappings**, and **workflow automation** within modern data stacks.
- Strong experience with **T-SQL** (DDL & DML, TCL, DCL) in Implementing & Developing Stored Procedures, Triggers, Nested Queries, Joins, Cursors, Views, User Defined Functions, Indexes, User Profiles, Relational Database Models, Creating & Updating tables.
- Extensive experience in developing applications that perform Data Processing tasks using **Teradata**, **Oracle**, **SQL Server**, and **MySQL database**.
- Led **Agile project management** initiatives, adeptly converting user stories into technical deliverables, managing stakeholder relations, spearheading process re-engineering, and overseeing change requests to ensure timely and efficient project delivery.
- Established strategic leadership to align development team efforts with business objectives, leveraging a robust understanding of disciplined **SDLC** processes and exceptional analytical and problem-solving capabilities.
- Played a pivotal role in the full project lifecycle, from requirements analysis to design, development, testing, deployment, and post-deployment support, utilizing **Agile** and **Waterfall** methodologies to optimize project outcomes.

EDUCATION:

- **Master of Science in Information Technology**, University Of Cincinnati, Cincinnati, Ohio, USA, Jan 2012 **-** Dec 2013
- **Bachelor of Technology in Information Technology**, Jawaharlal Nehru Technological University, Hyderabad, India, Jun 2007 – May 2011

TECHNICAL SKILLS:

Azure Technologies:	Azure Data factory (ADF), Azure Databricks, Azure Synapse Analytics, Azure Data Lake Storage Gen2, Azure Storage, Azure Event Hub, Azure IOT Hub, Azure SQL Database, Azure Data Catalog, Azure Logic Apps, Azure Functions, Azure Blob Storage, Azure DevOps, Azure Virtual Machine, Azure Active Directory, Azure Monitor, Azure Purview, Azure Cosmos DB, Azure Key Vault, Azure Repos, Azure HD Insight, Azure Kubernetes Services, Azure Stream Analytics, AWS (S3, Redshift, Lambda), Google Cloud (BigQuery, Cloud Functions), Monitoring & Logging (ELK Stack, Splunk, Datadog), IAM (Identity and Access Management), Role-Based Access Control.
Big Data Technologies:	HDFS, Map Reduce, YARN, Hive, MuleSoft, Informatica**,** SQOOP, Apache Spark, Kafka, Airflow, Spark Performance Tuning, Polybase, HQL, OOZIE, DataStage, Zookeeper, Pig, Flink, Spark Streaming,DBT Labs, Cloudera, Hortonworks, Databricks Delta, Apache NiFi, Snowflake, Azure Synapse, Amazon Redshift, dbt, Informatica PowerCenter, Talend, Matillion, SAP Data Integration, SAP BODS.
Databases:	MS SQL Server 2016/2014/2012, Azure SQL DB, Azure Synapse, MS Excel, Oracle 11g/12c, MS SQL, RDMS, Cassandra, MS Access, Cosmos DB, MongoDB, PostgreSQL, IBM DB2, Star Schema, Snowflake Schema, Dimensional Modelling, Snowflake, Partitioning, Clustering, Materialized Views, Kimball, Inmon Data Modeling.
Version Control & CI/CD Tools:	GIT , GitHub, Jenkins, Bitbucket, GitLab, Azure DevOps, Jenkins, Terraform, Kubernetes, Docker, Helm, GitHub Actions, YAML Pipelines, CloudFormation, Bicep,RBAC, IAM Policies, Security & Compliance.
Programming Languages:	Java, Python, PySpark, Shell Script, SQL, Scala, HiveQL, PL/SQL, T-SQL, T-SQL (Query Optimization), Pandas, Salesforce APIs (REST, SOAP), Apex.
IDE & Build Tools:	PyCharm, Eclipse, Visual Studio, SSIS, SSAS, SSRS, SSMS.
Visualization/Reporting Tools:	Tableau, Power BI, Alteryx, Ant, Maven, Terraform, Microsoft Fabric, QLIKVIEW.
Operating Systems:	Windows (XP/7/8/10), UNIX, LINUX, UBUNTU, CENTOS.
Methodology:	Apache Airflow, Prefect, Agile, Scrum, Azure Synapse Pipelines, Azure Data Factory Pipelines, Oozie, CI/CD Deployment Pipelines.

WORK EXPERIENCE:

Client: Vanguard **Jul 2021 - Current**
Location: Malvern , PA
Role: Sr. Azure Data Engineer
Responsibilities:

- Migrated 20TB+ of data from on-prem SFTP to Azure Data Lake (ADLS Gen2) using Azure Data Factory (ADF), reducing ETL processing time by 30% and improving data availability for analytics teams.
- Optimized Azure SQL Database performance by implementing bulk inserts, materialized views, and query parallelization, reducing execution time by 30% and improving data retrieval speeds for Azure-based analytics workloads.
- Designed and implemented Azure Synapse Analytics with Serverless SQL Pools, leveraging partitioning, indexing, and materialized views to reduce query execution times by 60%.
- Developed highly efficient ETL workflows in Azure Databricks using Delta Lake, Auto Loader, and Z-Ordering, improving data retrieval speeds for real-time Azure-based analytics and ensuring ACID compliance.
- Optimized Azure Blob Storage and ADLS hierarchical namespace for cost-efficient, high-throughput batch processing, reducing latency in Azure Synapse Analytics queries.
- Built scalable, high-performance data pipelines using Azure Databricks, Apache Spark, IBM DataStage, and Azure Delta Lake, enabling real-time financial market data processing and ensuring compliance with Azure governance standards.

- Worked with IBM DB2 databases to migrate legacy financial datasets to Azure Synapse Analytics, ensuring high availability and query performance optimization.
- Designed and developed ETL pipelines using IBM DataStage to extract, transform, and load structured and semi-structured financial data into Azure Data Warehouse environments.
- Implemented Slowly Changing Dimensions (SCD) Type 2 in Azure Delta Lake and IBM DataStage, ensuring accurate historical tracking of stock market transactions.
- Developed real-time data streaming solutions using Azure Event Hubs, Apache Kafka, and IBM MQ, enabling low-latency trade execution insights for high-frequency trading systems.
- Integrated Azure Synapse Pipelines for orchestrating end-to-end ETL processes, ensuring seamless data movement across Azure storage, Synapse SQL pools, and IBM DB2.
- Optimized Azure Databricks Spark workloads by implementing Azure partitioning, bucketing, caching, and Delta Caching, reducing data processing time by 40% and improving Azure resource utilization.
- Leveraged Adaptive Query Execution (AQE) in Azure Databricks to dynamically optimize Azure SQL joins, shuffle partitions, and Azure batch workloads, cutting Azure compute costs by 35%.
- Implemented enterprise-wide data governance using Unity Catalog, IBM DataStage, and Azure Purview, ensuring GDPR, SOC 2, and SEC compliance while providing end-to-end data lineage tracking.
- Enforced role-based access control (RBAC) with Azure Entra ID (formerly Azure AD) to manage fine-grained permissions for financial datasets, ensuring data privacy and security compliance.
- Automated user provisioning and de-provisioning across Azure resources using Azure Entra ID and IBM Security Access Manager, streamlining access management and ensuring adherence to regulatory requirements.
- Secured sensitive credentials, API keys, and encryption secrets using Azure Key Vault, integrating with Azure Databricks, IBM DataStage & ADF pipelines for seamless secure access control.
- Established encryption at rest and in transit using Azure Defender for SQL & Azure ADLS Secure Transfer, preventing data breaches and unauthorized access to Azure storage.
- Designed and deployed end-to-end CI/CD pipelines in Azure DevOps, automating Azure infrastructure provisioning and ETL pipeline deployments using YAML pipelines, Terraform, and IBM UrbanCode Deploy.
- Developed and optimized ETL workflows using IBM DataStage, Matillion, Apache NiFi, and Azure Data Factory to process large financial datasets in Snowflake and Azure Synapse Analytics.
- Implemented DBT (Data Build Tool) for automated data transformations, source-to-target mapping, and modular SQL development, enhancing data model maintainability and consistency across ETL pipelines.
- Developed Dimensional Modeling (Star Schema & Snowflake Schema) in Azure Synapse Analytics & IBM DB2, improving OLAP performance for complex financial analysis.
- Built OLAP cubes using Azure Synapse & IBM Cognos Analytics, enabling faster trend forecasting, portfolio risk analysis, and predictive market insights.
- Designed and implemented real-time Power BI and Tableau visualizations using Microsoft Fabric & Direct Lake Mode, enabling low-latency financial reporting and analytics.
- Integrated MLflow with Azure Databricks to implement real-time sentiment analysis on financial news, improving trade decision-making using AI-based predictive insights.
- Optimized Azure-based ETL workflows & governance policies, improving data accuracy, reducing compliance risks, and enhancing regulatory reporting.
- Implemented robust RBAC and IAM policies in Azure Entra ID, ensuring fine-grained access control for sensitive financial data and adhering to stringent security compliance standards.
- Enabled real-time risk modeling & AI-driven predictive analytics, empowering investment teams to make smarter, data-driven trading decisions.

Environment: Azure Databricks, Azure Data Factory (ADF), Azure Synapse Analytics, Azure SQL Database, Azure Cosmos DB, Azure Blob Storage, Azure Data Lake Storage (ADLS Gen2), Delta Lake, Azure Stream Analytics, Logic Apps, Azure HDInsight, Azure Event Hubs, Azure Functions, Azure Key Vault, Azure Monitor, Azure DevOps, Terraform, ARM Templates, Azure Purview, Unity Catalog, NoSQL, DAG, HDFS, Apache Spark, Spark Structured Streaming, Spark SQL, Hive, T-SQL, PostgreSQL, Snowflake, Python, Scala, PySpark, Spark Performance Tuning, Adaptive Query Execution (AQE), DAX, PowerApps, Shell Scripting, Apache Kafka, Confluent Kafka, Airflow, Microsoft Fabric, Power BI, Control-M Scheduler, Alteryx, Tableau.

Client: Bank of America **Mar 2018 - Jun 2021**
Location: Charlotte, NC
Role: Azure Data Engineer

Responsibilities:

- Designed and implemented end-to-end data pipelines using Azure Databricks, Spark, DBT models, and Azure Data Factory to transform and aggregate large scale ATM and branch network data for optimization and predictive analytics.
- Designed and optimized Azure Synapse Analytics solutions to enhance data warehousing capabilities, integrating Synapse Pipelines, Dedicated SQL Pools, and Serverless SQL Pools to improve analytical query performance by 40%.
- Implemented data partitioning, indexing, and materialized views in Azure Synapse to optimize query performance and ensure seamless data integration with Snowflake, Databricks, and Azure Data Factory, reducing ETL processing time by 30%.
- Optimized data transformation workflows, resulting in a 40% reduction in processing time for large datasets, handling data sizes of up to 100 terabytes.
- Developed ETL pipelines integrating data from SAP systems while implementing SAP BODS into Azure Synapse, ensuring seamless data transformation for financial analysis and risk reporting
- Engineered Azure Data Factory pipelines to automate data ingestion from multiple sources, processing up to 100 terabytes of structured and unstructured data daily for financial and ATM analytics.
- Implemented Azure Active Directory (Azure AD) for secure authentication and authorization of data pipelines and services, ensuring role-based access control for sensitive financial data within Azure Synapse and Databricks.
- Integrated Azure AD with Azure Data Lake Storage (ADLS) and other Azure services to manage identity and access, streamlining user authentication for data engineers and analysts accessing critical datasets and reports.
- Developed real-time data streaming solutions with Azure Event Hubs, Kafka, and Azure Functions, enabling predictive maintenance and cash replenishment forecasting for ATMs and bank branches.
- Optimized data lake storage and data warehouse architectures using Azure Data Lake Storage (ADLS) and Snowflake, reducing data retrieval times by 40% and improving query performance by 25%.
- Implemented robust ETL pipelines using Apache Airflow, Talend, and Snowflake, automating data transformations and improving data integrity and governance across financial datasets.
- Leveraged Azure Blob Storage and compression techniques for efficient data storage and encryption, optimizing cloud storage costs and ensuring compliance with regulatory standards (SOX, CCAR, GDPR).
- Developed and maintained Spark-based ETL solutions in Databricks, improving data processing efficiency for large-scale datasets, reducing query latencies, and enhancing fraud detection capabilities.
- Designed HIVE and Spark SQL transformations to improve data aggregation and analytical processing, reducing reporting latency and enabling faster decision-making for branch and ATM optimizations.
- Created and managed responsive, resilient, and scalable data architectures using Azure SQL DW, NoSQL DB, and HDInsight/Databricks, ensuring high availability for banking operations.
- Implemented streaming analytics pipelines using Spark Streaming and Kafka, enhancing real-time monitoring of ATM cash withdrawal trends and enabling proactive fraud detection mechanisms.
- Developed Python and Scala-based ETL frameworks, transforming structured and unstructured data into actionable insights for business intelligence and predictive analytics in ATM and branch networks.
- Designed and implemented data partitioning strategies and data retention policies in ADLS, significantly optimizing storage performance and reducing cloud costs.
- Engineered CI/CD workflows for deploying data pipelines using GCP Cloud Build, Cloud SDK, Jenkins, and Terraform, ensuring seamless deployment and versioning of ETL pipelines.
- Created real-time and batch reporting dashboards using Power BI, Google Data Studio, and BigQuery, enabling actionable insights into ATM cash demand forecasting and network efficiency.
- Applied advanced performance tuning techniques for Spark and Hive queries, reducing job execution times and enhancing efficiency for high-volume transaction processing.
- Developed automated monitoring and alerting mechanisms in Splunk and Azure Monitor, ensuring proactive issue resolution and maintaining 99.99% uptime for critical banking data pipelines.
- Utilized DBT models and Apache Oozie for workflow orchestration, automating the execution of Spark, Hive, and MapReduce jobs, ensuring seamless and optimized data processing workflows.
- Led the migration of legacy Hadoop data pipelines to Snowflake, implementing schema optimization techniques, improving query execution times by over 30% for analytical workloads.
- Participated actively in Agile Scrum methodologies, contributing to sprint planning, backlog grooming, and daily stand-ups to ensure efficient development cycles and on-time project delivery.

Environment: Azure Databricks, Azure Synapse, Data Factory, Snowflake, Kafka, Logic Apps, Functional Apps, Azure Event Hubs, Azure Blob Storage, MS SQL, Oracle, MongoDB, DBT, HDFS, MapReduce, YARN, Spark, Hive, SQL, Python, Scala, PySpark, Apache Airflow, Shell Scripting, GIT, Jenkins, ADF Pipeline, Power BI, Google Data Studio, GCP (BigQuery, Cloud SDK, IAM, Cloud Build, Security Command Center).

Client: Manulife. **Dec 2015 - Feb 2018**
Location: New York, NY
Role: Big Data Developer
Responsibilities:

- Designed and implemented an ETL framework using Sqoop, Pig, and Hive in the retail domain, facilitating the extraction and integration of data from diverse sources for seamless consumption.
- Designed and implemented scalable data lake architecture, incorporating HDFS, to efficiently store and manage vast volumes of data.
- Processed data stored in Hadoop Distributed File System (HDFS) and created external tables using Hive, developing efficient scripts for table ingestion and reparation for reusability across the project
- Managed and optimized Hadoop clusters, dynamically scaling resources based on workload demands for efficient data processing.
- Implemented robust data security measures, including encryption, authentication, and authorization, to protect sensitive data in Hadoop environments.
- Developed robust ETL jobs using Spark and Scala, enabling the seamless migration of data from Oracle databases to new MySQL tables in the retail domain
- Leveraged Spark's capabilities, including RDDs, Data Frames, and Spark SQL, along with Spark-Cassandra Connector APIs, for various data-related tasks, such as data migration and generating business reports
- Designed and implemented a real-time sales analytics application using Spark Streaming, enabling instant insights into retail sales data
- Conducted in-depth analysis of source data, performed necessary data type modifications, and effectively utilized Excel sheets, flat files, and CSV files to generate ad-hoc reports in Power BI for comprehensive retail data analysis
- Analyzed SQL scripts and designed solutions using PySpark, ensuring efficient data processing and manipulation in the retail domain. Developed parameterized SQL queries to dynamically adjust data transformations based on reporting needs.
- Extracted data from various data sources into HDFS using Sqoop, enabling seamless integration and utilization of diverse retail datasets. Implemented error handling and logging mechanisms to track ingestion failures and ensure data integrity.
- Managed data import from multiple sources, performed transformations using Hive and MapReduce, and loaded processed data into HDFS. Designed optimized Hive queries leveraging bucketing, partitioning, and indexing to improve performance.
- Designed and implemented Star and Snowflake Schemas for efficient data modeling and reporting within the Big Data environment. Created dimension tables and fact tables to support analytical queries and BI tools.
- Extracted data from MySQL into HDFS using Sqoop and optimized ingestion processes using parallel data transfers. Employed Zookeeper for centralized service management, ensuring configuration synchronization and distributed coordination in Big Data applications.
- Implemented automation for deployments using YAML scripts, streamlining build and release processes. Integrated CI/CD pipelines for automated deployment of Hadoop jobs and Spark-based transformations.
- Optimized query performance in Hive using bucketing and partitioning techniques, significantly reducing query execution time for large-scale data sets. Developed materialized views and precomputed aggregations to enhance reporting efficiency.
- Collaborated with Apache Hive, Apache Pig, HBase, Apache Spark, Zookeeper, Flume, Kafka, and Sqoop, ensuring smooth and scalable end-to-end data ingestion, transformation, and processing workflows for high-volume datasets.
- Implemented data classification algorithms using MapReduce design patterns, enabling automated categorization of customer transaction data. Developed machine learning-based categorization workflows for predictive analytics.
- Extensively worked on creating combiners, partitioning, and distributed cache, improving the performance of MapReduce jobs. Designed optimized shuffle and sort operations to reduce data movement and improve job execution efficiency.
- Implemented Git and GitHub version control, ensuring structured repository management, commit tracking, and seamless collaboration across teams. Developed branching strategies to streamline ETL code deployment and updates.

Environment: Hadoop components - HDFS, MapReduce, Hive, spark, PySpark,DBT models, Sqoop, Spark SQL, Shell script, Cassandra, YAML, ETL, Data Warehouse, Zookeeper, Flume, Yarn, HBase.

Client: Kroger. **Jan 2014 - Nov 2015**
Location: Cincinnati, OH
Role: ETL Developer
Responsibilities:

- Automated report generation and Cube refresh processes through creating SSIS jobs, ensuring timely and accurate delivery of critical information to stakeholders.
- Developed complex stored procedures, efficient triggers, and necessary functions to optimize SQL Server performance.
- Applied extensive experience in the analysis, design, development, implementation, enhancement, and support of BI applications within the data warehousing ETL OLAP environment as a data warehouse consultant.
- Monitored and fine-tuned SQL Server performance, applying best practices to ensure optimal database performance.
- Demonstrated expertise in designing ETL data flows using SSIS and creating mappings and workflows for extracting data from SQL Server. Executed data migration and transformation from Access/Excel sheets using SQL Server SSIS.
- Proficiently implemented dimensional data modeling for Data Mart design, identifying facts and dimensions and developing fact tables and dimension tables using Slowly Changing Dimensions (SCD) techniques.
- Applied skills in error and event handling techniques, ensuring reliable and robust ETL processes through precedence constraints, breakpoints, checkpoints, and logging.
- Successfully built cubes and dimensions with different architectures and data sources for business intelligence purposes, including writing MDX scripting.
- Wrote SQL queries, including DDL, DML, and diverse database objects (indexes, triggers, CTEs, views, stored procedures, functions, and packages) for data manipulation and retrieval.
- For better performance, have incorporated various stored procedures, merges, view, and various window functions like rank(), dense rank(), etc.
- Used various transformations like Filter, Expression, Sequence Generator, Update Strategy, Joiner, Stored Procedure, and Union to develop robust mappings in the Informatica Designer.
- Implemented compliance and auditing features within Informatica Enterprise Data Catalog, ensuring adherence to regulatory requirements and providing transparency for audit purposes within the data warehouse environment.
- Proficient in developing SSAS cubes, implementing aggregations, defining KPIs, managing measures, partitioning cubes, and creating data mining models. Deployed and processed SSAS objects.
- Demonstrated expertise in creating ad hoc reports and reports with complex formulas, utilizing querying capabilities of the database for business intelligence purposes.
- Implemented advanced reporting functionalities in Power BI, including Drill-through and Drill-down reports, interactive Drop-down menus, data sorting capabilities, and subtotals for enhanced data analysis.
- Applied Data warehousing techniques to develop a comprehensive Data Mart, serving as a reliable data source for downstream reporting. Developed a User Access Tool empowering users to create ad-hoc reports within the proposed Cube by deploying SSIS Packages to production, achieving environment independence.
- Utilized SQL Server Reporting Services (SSRS) to author, manage, and deliver comprehensive reports in print and web-based formats using robust stored procedures and triggers for data consistency and integrity during data entry operations.
- Developed robust stored procedures and triggers for data consistency and integrity during data entry operations.
- Utilized Informatica Enterprise Data Catalog for data discovery, profiling, and analysis, identifying data quality issues and opportunities for performance optimization within the data warehouse.
- Mapped sources to targets using tools like Business Objects Data Services/BODI and designed ETL code to load and transform source data into an SQL database.
- Created, launched, and scheduled tasks/sessions, configured email notifications, and set up tasks for scheduled loads using Power Center Server Manager.
- Developed Static and Dynamic Parameter Files, ensuring reusability and streamlined database connection management across Development, Testing, and Production environments. Designed configurations to support multi-environment deployment strategies.

- Utilized Visual SourceSafe (VSS) for version control, maintaining code repositories, enforcing best practices in source code management, and tracking historical changes in ETL workflows. Effectively managed project tracking using Trello for task prioritization and sprint planning.

Environment: MS Access, MS SQL Server 2012, SSMS, SSIS, SSRS, SSAS, Slowly Changing Dimensions (SCD), triggers, Informatica Power Center 9.1.0, Oracle 10g, SQL Server 2005, SQL Loader, Star and Snowflake Schemas, Share point, Power BI, Visual SourceSafe, Trello, Git